

Knowledge Machine v3

Canonical Engineering Notes & Master Constitution

1. System Identity & Purpose

Knowledge Machine v3 is a deterministic, offline, computer-vision-based knowledge extraction engine. It converts long educational videos into a minimal, information-dense set of visual states that reflect how a disciplined human student would take notes. It is not an AI summarizer, not an OCR-first system, and not a probabilistic tool. Its primary function is to fossilize human explanations into permanent, reviewable memory without hallucination or loss.

2. Core Objectives Timeline

- 1 Objective 1: Zero-Loss Capture — Never miss semantic information (ACHIEVED).
- 2 Objective 2: Human-Aligned Deduplication — Reduce noise while keeping meaning (ACHIEVED).
- 3 Objective 3: Semantic Sweet Spot — No AI/OCR in decision logic (ACHIEVED).
- 4 Objective 4: Adaptive Sensitivity — Code vs theory vs talking head (ACHIEVED).
- 5 Objective 5: Offline, Free, Deterministic Operation (ACHIEVED).
- 6 Objective 6: Time Compression — Hours to minutes (ACHIEVED).
- 7 Objective 7: AI-Ready Outputs — Clean visual corpus (PARTIAL, metadata pending).

3. Two-Phase Architecture (Locked)

Phase separation is sacred and irreversible. Phase 1: Paranoid Capture Implemented in `smart_frame_selector.py`. Purpose: Never miss meaning. Allows noise but forbids blindness. Uses FPS sampling, content masking, pixel deltas, temporal plateaus, flash buffers, optical-flow scroll detection, and scroll latching. Phase 2: Ruthless Consolidation Implemented in `screenshot_consolidator.py`. Purpose: Remove redundancy without merging meaning. Uses SSIM ≥ 0.95 , ink-density estimation, superset replacement rules, scroll-aware guards, and resolution safety.

4. The Semantic Sweet Spot (Invariant)

A frame may only replace another frame if it is provably a semantic superset. This invariant prevents loss, hallucination, and accidental merging. It is enforced through structural similarity, relative ink growth, and scroll isolation.

5. File Map & Roles

- 1 `master_pipeline.py` — Orchestrator; runs the entire pipeline.
- 2 `extract_all.py` — Infrastructure; downloads video and prepares workspace.
- 3 `smart_frame_selector.py` — Phase 1 watcher; paranoid capture logic.
- 4 `screenshot_consolidator.py` — Phase 2 editor; semantic deduplication.
- 5 `ocr_processor.py` — Optional post-process indexer; never used for decisions.
- 6 Debug, safety, and legacy files — Historical and diagnostic, not production.

6. Known Failure Modes (Solved)

- 1 Typing ghosts → Stability plateau.
- 2 Cursor blink corruption → Relative ink margin.
- 3 Scrolling blindness → Optical flow + scroll flag.
- 4 Micro-flashes → Ring buffer capture.
- 5 UI noise → Central masking.
- 6 Over-deduplication → SSIM + ink logic.

7. Explicitly Forbidden Actions

OCR or AI inside capture or consolidation, lowering SSIM thresholds for technical content, merging scroll states, reducing resolution, guessing missing information, or skipping safety checks. Violation of these breaks system correctness.

8. What Is Intentionally Missing

Session-level metadata, timeline JSON mapping, and failure logs are not yet built. These are non-destructive Phase 3 extensions and do not affect capture correctness. AI is allowed only downstream of consolidation.

9. System Contract for Downstream AI

Downstream AI may summarize, explain, index, or structure content strictly based on visible slides. If information is not present, the AI must explicitly state so. Inference beyond visual evidence is forbidden.

10. Final Assessment

Knowledge Machine v3 is 85–90% complete. The remaining work is leverage: metadata, packaging, and downstream intelligence. The hard engineering problems are solved. The system is stable, finished, and future-proof.